

Math 161

Problem set 3 solutions

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1. (a) We proved that least upper bounds are unique in class; the following is the same argument with inequalities reversed.

We have $x \leq x'$ because x is a lower bound and x' is a greatest lower bound. Similarly, we have $x' \leq x$ because x' is a lower bound and x is a greatest lower bound. Thus $x = x'$.

- (b) Since $A \neq \emptyset$, there exists $x \in A$. Thus $-x \in -A$ and hence $-A \neq \emptyset$. To see that $-A$ is bounded above, we use the hypothesis that A is bounded below. The latter means that there exists $y \in \mathbb{R}$ such that $y \leq x$ for all $x \in A$. But then $-x \leq -y$ for all $y \in A$, which shows that $-y$ is an upper bound for $-A$.

Thus $\sup(-A)$ exists by the least upper bound property of $\mathbb{R}$. To see that $-\sup(-A) = \inf A$, we first show that $-\sup(-A)$ is a lower bound for A . Now $\sup(-A)$ is an upper bound for $-A$, i.e. $\sup(-A) \geq -x$ for all $x \in A$. This implies that $-\sup(-A) \leq x$ for all $x \in A$, which exactly means that $-\sup(-A)$ is a lower bound for A . Finally, to see that $-\sup(-A)$ is the greatest lower bound for A , suppose that $y \in \mathbb{R}$ is a lower bound for A , i.e. $y \leq x$ for all $x \in A$. Then we have $-x \leq -y$ for all $x \in A$, i.e. $-y$ is an upper bound for $-A$. Since $\sup(-A)$ is the least upper bound for $-A$, we therefore have $\sup(-A) \leq -y$. Thus $y \leq -\sup(-A)$ as needed.

- (c) The hypothesis that A is bounded below means that A admits a lower bound y . Thus we have $y \in B$ and hence $B \neq \emptyset$. Since $A \neq \emptyset$ there exists $x \in A$, and by the definition of B we have $x \geq y$ for all $y \in B$, i.e. x is an upper bound for B . So B is bounded above.

Finally, by definition $\inf A$ is the greatest element of B . We saw in class that the greatest element of a set is automatically its least upper bound, so $\inf A = \sup B$.

2. (a) Denote the set in question by A . We observe that A has greatest element (hence supremum) 1, because $n \geq 1$ and hence $\frac{1}{n} \leq 1$ for all $n \in \mathbb{N}$. On the other hand, we claim that $\inf A = 0$, which means that A does not have a least element because $0 \notin A$. For this, first note that $0 < \frac{1}{n}$ for all $n \in \mathbb{N}$, and hence 0 is a lower bound for A . If $x > 0$, then by the (corollary to the) Archimedean property of $\mathbb{R}$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < x$. In particular x is not a lower bound for A , so any lower bound $y \in \mathbb{R}$ for A must satisfy $y \leq 0$. Thus $\inf A = 0$ as desired.
- (b) This set has greatest element 1 and least element -1 . Namely, we have

$$1 \geq \frac{1}{n} > 0 > -\frac{1}{n}$$

and

$$-1 \leq -\frac{1}{n} < 0 < \frac{1}{n}$$

for all $n \in \mathbb{N}$.

- (c) This set has greatest element 1 and least element 0. This is similar to, but easier than, part (a): just observe that

$$1 \geq \frac{1}{n} > 0$$

for all $n \in \mathbb{N}$.

- (d) Let A denote the set in question. We claim that $\frac{3}{2}$ is the greatest element, and that $\inf A = -1$. Indeed, for even $n \in \mathbb{N}$ we have

$$\frac{3}{2} = \frac{1}{2} + (-1)^2 \geq \frac{1}{n} + (-1)^2 = \frac{1}{n} + (-1)^n,$$

where for the inequality we used that $n \geq 2$ and hence $\frac{1}{n} \leq \frac{1}{2}$ for n even. For odd n we have that

$$\frac{1}{n} + (-1)^n = \frac{1}{n} - 1 \leq 0 < \frac{3}{2},$$

so $\frac{3}{2}$ is indeed the greatest element of A .

As for $\inf A = -1$, first observe that for odd $n \in \mathbb{N}$ we have

$$-1 < \frac{1}{n} - 1 = \frac{1}{n} + (-1)^n,$$

and

$$-1 < 1 < \frac{1}{n} + 1 = \frac{1}{n} + (-1)^n$$

for even n . Thus -1 is a lower bound for A . To see that it is the least upper bound, suppose that $x \in \mathbb{R}$ satisfies $x > -1$. Then $x + 1 > 0$, so by the Archimedean property of $\mathbb{R}$ there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < x + 1$ and hence $\frac{1}{n} - 1 < x$. Certainly we can choose this n to be odd: if n is even, simply add 1 to it and use the fact that $\frac{1}{n+1} < \frac{1}{n}$. Then we have

$$\frac{1}{n} + (-1)^n = \frac{1}{n} - 1 < x,$$

i.e. x is not a lower bound for A . Thus we have shown that any lower bound $y \in \mathbb{R}$ for A satisfies $y \leq -1$, so $\inf A = -1$ as claimed.

3. (a) There are a couple of ways to prove that there exist a largest $\ell \in \mathbb{Z}$ such that $\ell \leq x$. One way is to first apply the Archimedean property of $\mathbb{R}$ to find $N \in \mathbb{N}$ such that $N > x$. Now consider the set

$$A := \{n \in \mathbb{N} \mid N - n \leq x\} \subset \mathbb{N}.$$

We claim that $A \neq \emptyset$. Namely, another application of the Archimedean property produces $M \in \mathbb{N}$ such that $M \geq N - x$, which means that $N - M \leq x$ or equivalently $M \in A$. Since $A \subset \mathbb{N}$ and $A \neq \emptyset$, the well-ordering principle implies that A admits a least element n , and we can put $\ell := N - n$. By definition $\ell \leq x$, so suppose $m \in \mathbb{Z}$ also satisfies $m \leq x$. Then $N - m \in A$, because

$$N - (N - m) = m \leq x,$$

and hence $N - m \geq n$ because n is the least element of A . This means that $\ell = N - n \geq m$, so ℓ is the largest integer with the property that $\ell \leq x$, as claimed.

Here is a sketch of another approach to the existence of ℓ : consider the set

$$B := \{n \in \mathbb{Z} \mid n \leq x\}.$$

One shows that B is nonempty using the Archimedean property, and B is certainly bounded above by x , so $\sup B$ exists in $\mathbb{R}$. One then shows somehow that $\sup B \in \mathbb{Z}$, so we can put $\ell := \sup B$.

In any case, we have now proved the existence of ℓ . Since $\ell + 1 > \ell$ and $\ell + 1 \in \mathbb{Z}$, we must have $\ell + 1 > x$. On the other hand, we have

$$\ell + 1 \leq x + 1 < x + y - x = y.$$

Thus $k := \ell + 1 \in \mathbb{Z}$ satisfies $x < k < y$, as needed.

- (b) Suppose that $x, y \in \mathbb{R}$ satisfy $x < y$, i.e. $y - x > 0$. By the Archimedean property, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < y - x$. This implies that $1 < ny - nx$, so by part (a) there exists $k \in \mathbb{Z}$ such that $nx < k < ny$. Finally, divide by n to obtain $x < \frac{k}{n} < y$, and since $\frac{k}{n} \in \mathbb{Q}$ we are done.
4. By the least upper bound property of $\mathbb{R}$, if we can prove that A is bounded above then it will follow that $\sup A$ exists (we have $A \neq \emptyset$ already by (iii)). So suppose for the sake of contradiction that A is not bounded above. This means that for any $y \in \mathbb{Q}$, there exists $x \in A$ such that $y < x$. But then $y \in A$ by (iv), so we have shown that $A = \mathbb{Q}$, contradicting (ii).

To see the inclusion

$$A \subset \{x \in \mathbb{Q} \mid x < \sup A\},$$

fix $x \in A$, and note that $x \in \mathbb{Q}$ by (i). We must also show that $x < \sup A$, so assume for the sake of contradiction that $x \geq \sup A$. By (v) there exists $z \in A$ such that $x < z$, but then $\sup A \leq x < z$, contradicting the fact that $\sup A$ is an upper bound for A .

As for the inclusion

$$\{x \in \mathbb{Q} \mid x < \sup A\} \subset A,$$

we must show that if $x \in \mathbb{Q}$ satisfies $x < \sup A$, then $x \in A$. First observe that $x < \sup A$ implies that there exists $x' \in A$ such that $x < x'$. Indeed, otherwise x would be an upper bound for A , contradicting that $\sup A$ is the least upper bound of A . But now (iv) implies that $x \in A$.

5. (a) The hypothesis that X and Y are countably infinite means that we can enumerate their elements in a sequence

$$X = \{x_1, x_2, \dots\} \text{ and } Y = \{y_1, y_2, \dots\},$$

where each sequence contains no repetitions, i.e. $x_i \neq x_j$ and $y_i \neq y_j$ for $i \neq j$. Now we can arrange the elements of $X \times Y$ in an infinite table:

(x_1, y_1)	(x_2, y_1)	(x_3, y_1)	
(x_1, y_2)	(x_2, y_2)	(x_3, y_2)	$\dots$
(x_1, y_3)	(x_2, y_3)	(x_3, y_3)	
	$\vdots$		

This allows us to enumerate the elements of $X \times Y$ by making a zigzag, starting in the upper left:

$$X \times Y = \{(x_1, y_1), (x_2, y_1), (x_1, y_2), (x_1, y_3), (x_2, y_2), (x_3, y_1), (x_4, y_1), (x_3, y_2), \\ (x_2, y_3), (x_1, y_4), (x_1, y_5), \dots\}$$

(one can write a formula for the n^{th} element of this sequence, but it is quite tedious). In this way we will obtain all elements of $X \times Y$ exactly once.

- (b) First note that $\mathbb{Z}$ is countably infinite, using e.g. the sequence

$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \dots\}.$$

Similarly, the subset

$$\mathbb{Z} \setminus \{0\} = \{1, -1, 2, -2, 3, -3, \dots\}$$

is countably infinite. Since any element of $\mathbb{Q}$ can be represented as a fraction $\frac{a}{b}$ where $a, b \in \mathbb{Z}$ and $b \neq 0$, the map

$$\begin{aligned} \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) &\longrightarrow \mathbb{Q} \\ (a, b) &\mapsto \frac{a}{b} \end{aligned}$$

is surjective (note that for any $x \in \mathbb{Q}$, the preimage of $\{x\}$ under this map is the equivalence class of ordered pairs which represent x as a fraction). By part (a) the set $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ is countably infinite, so it follows that $\mathbb{Q}$ is finite or countably infinite. Certainly $\mathbb{Q}$ is infinite, since e.g. it contains the infinite subset $\mathbb{N} \subset \mathbb{Q}$. Thus $\mathbb{Q}$ is countably infinite.